

Stakeholder perspectives on ecosystem services in agricultural landscapes: A case study in the North China Plain

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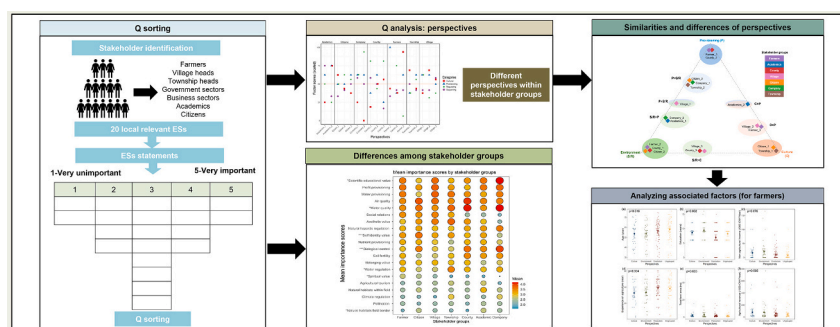
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HIGHLIGHTS

- Landscape redesign may improve ecosystem services (ESs) but implementation requires stakeholder support.
- Q methodology was used to characterize stakeholder perspectives on ESs in Quzhou county, China.
- Stakeholders prioritized provisioning services but also valued environmental quality and cultural values.
- Diverse stakeholders' perspectives highlight the potential trade-offs and decision-making challenges for policy.

GRAPHICAL ABSTRACT



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ABSTRACT

CONTEXT: Intensive agriculture is under pressure from changing demands from society, prompting the need to redesign agricultural landscapes to provide multiple ecosystem services (ESs). However, implementation of changed practices requires positive engagement from stakeholders. Therefore, their perspective on ecosystem services needs to be known.

OBJECTIVE: This study investigates stakeholders' perspectives on multiple ESs in Quzhou County, an area in the North China Plain used for intensified cereal production. We aim to elucidate perspectives within and across diverse stakeholder groups (farmers, companies, citizens, academics, village and township heads, and county government staff).

METHODS: Employing the Q methodology, we identified differences in perspectives within stakeholder groups and we compared the similarities and differences of those perspectives across stakeholder groups. We also investigated how farmers' personal and household characteristics were related to the perspectives they held.

RESULTS AND CONCLUSIONS: Significant differences in preference emerged among stakeholder groups. Academics assigned higher importance to regulating and supporting services than other stakeholder groups and companies assigned less importance to cultural services. We identified 18 distinct perspectives across seven

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stakeholder groups. These perspectives showed a combination of preferences for at least two different ES categories. Most of the perspectives prioritize provisioning services whereas only few perspectives prioritize supporting services.

SIGNIFICANCE: This study exemplifies a bottom-up approach for systematically analyzing stakeholder perspectives on the relative importance of ESs derived from agricultural landscapes. The revealed differences and complexity of stakeholder perspectives can inform decision-making on the redesign of agricultural landscapes with stakeholder engagement. Recognizing areas of consensus and conflict can guide efforts to promote agro-ecologically sound practices and policies.

1. Introduction

Intensive agriculture needs transformation towards greater sustainability worldwide, which may require a redesign of agricultural landscapes (Jackson et al., 2007; Landis, 2017). Agricultural landscapes provide multiple ecosystem services (ESs) to a range of stakeholders (O’Farrell and Anderson, 2010). An unbalanced ESs delivery leads to compromised sustainability and biodiversity loss (Foley, 2005), potentially leaving some people’s demand for ESs unmet. It is necessary to assess the importance of ESs for stakeholders when adjusting the management of a landscape to rebalance the ESs supply. The importance of ESs for stakeholders is strongly shaped by the socio-economic and cultural context and by their (subjective) values, which mediate between the benefits derived from services and the processes of governance that steer agroecosystem management decisions (Breynne et al., 2021).

The value of ESs refers to economic (monetary) values, but also non-monetary, socio-cultural values (MEA, 2005). Economic values highlight the importance of certain ESs by assigning them high monetary worth, which helps raise awareness and communicate their significance to policymakers (de Groot et al., 2012). Socio-cultural values express people’s opinions on the non-monetary importance of ESs and are reflected in stakeholder preferences for the multiple ESs that are provided by an agroecosystem (see Table 1 for definitions). Insight into stakeholder preferences can firstly aid policymakers in understanding how changes in landscape management are perceived by different stakeholders. Secondly, it informs decision-making in situations where trade-offs among various (categories of) performance indicators occur (Scholte et al., 2015). More inclusive and equitable decision-making is facilitated by the integration of socio-cultural values with ecological and monetary assessment. Contrasting perspectives on these diverse values may exist and should be considered, in particular when stakeholders are engaged in the assessment process (Martín-López et al., 2012).

The degree to which the demand and supply of multiple ESs can be balanced within a society or community depends on the heterogeneity of demands, the potential changes that can be realized in the supply of the ESs and the extent to which spatial distances or temporal lags hamper the timely supply of demanded ESs (Mouchet et al., 2014). These considerations should also inform decision-making and policy formulation (Bulkeley and Mol, 2003). Here we focus only on the heterogeneity on the demand side. Different stakeholders that are active or have an interest in certain landscapes could have different preferences for ESs depending on their roles, demographic characteristics, and other

attributes (Smith and Sullivan, 2014). While farmers might prefer provisioning services for their well-being, urban visitors might show a preference for cultural services (Zoderer et al., 2019), while regional development programs and researchers would tend to assign greater importance to regulating services to support the sustainable use of the landscape (Peter et al., 2022). Within a certain stakeholder group, there may also be different perspectives among individuals (Teixeira et al., 2018). However, few studies have analyzed the heterogeneity of preferences within stakeholder groups (Bernues et al., 2019).

Different methodologies could be applied to assess socio-cultural values related to ESs, including expert-based approaches, focus group discussions, interviews and questionnaire surveys (Chen et al., 2017; Iniesta-Arandia et al., 2014; Lamarque et al., 2011; Page and Bellotti, 2015; Smith and Sullivan, 2014). However, as found in previous studies and in our own pilot survey, when respondents are asked about the importance of different ESs, nearly all respondents state that most ESs are important (Horne et al., 2005). However, not all ESs can be provided maximally at the same time. Trade-offs among ESs make it necessary to prioritize ESs in decision-making and planning processes. Hence there is a need for assessment approaches that give insight into the preferences of ESs among stakeholders.

Q methodology is a semi-quantitative method for eliciting and systematically presenting stakeholder perspectives. It originates from psychology and has also been used for many years in policy science and public health (McKeown and Thomas, 2013; Watts and Stenner, 2005; Valenta and Wigger, 1997). It aims to reveal the way individuals think about a specific subject (Watts and Stenner, 2005) and assesses if groups of individuals perceive a subject similarly (Stephenson, 1953). Conducting Q methodology involves generating a set of statements that represent the various facets of a discourse or problem, which are then sorted by participants by the degree of agreement with the statements. The sorting result is statistically analyzed to identify common sorting patterns among participants (Watts and Stenner, 2005). In this paper, we refer to typical sorting patterns as perspectives. Perspectives thus represent typical patterns of how stakeholders think about the relative importance of different ESs. Q sorting embedded in Q methodology encourages participants to sort the items using deliberate thinking.

Q methodology has gradually gained credence in environmental and agricultural science to assess the subjective values of stakeholders when designing management and policymaking (Jiren et al., 2023; Rietveld et al., 2023; Swagemakers et al., 2017; Timler et al., 2023; Vaas et al., 2019; van Dijk et al., 2022). There are however only few applications of Q methodology to capturing stakeholders’ perspectives on the importance of ESs in agricultural landscapes (Ciftcioglu, 2021) compared to other ecosystems, e.g., nature reserves (Mahlalela et al., 2022), national forests (Armatas et al., 2018), aquaculture (Pethiyagoda et al., 2024), and urban parks (Buchel and Frantzeskaki, 2015).

The North China Plain is one of the largest crop production areas in China and is characterized by simplified agricultural landscapes with a dominance of wheat, maize and cotton (Xu et al., 2024). As a major breadbasket of China, the region is geared towards enhancing provisioning services (i.e. production of food, feed and other commodities), and generation of economic value. Regions with intensive production are often characterized by sustainability issues such as declining groundwater tables (Zhao et al., 2017), greenhouse gas emissions (Zhao

Table 1
Glossary of most important terms in this study.

Terms	Definitions
Value	The importance and necessity of objects and actions (Harrington et al., 2010) described in monetary or non-monetary terms (MEA, 2005).
Importance	Indication to what extent and how a service or its associated benefits matter (including non-monetary terms benefits) for a person or a group of persons (Breynne et al., 2021).
Preference	Preference refers to the importance attributed to one ES relative to another one (IPBES, 2019).
Perspective	A perspective in this case refers to the combination of preferences towards multiple (bundles of) ecosystem services that are shared by a group of stakeholders.

et al., 2021) and biodiversity loss (Mu et al., 2022), i.e. a lack of non-provisioning services (regulating, supporting, and cultural services) (Cheng et al., 2023). Those non-provisioning services have been less emphasized in this region and may need revitalization by policy for the sake of long-term sustainability of the agroecosystem. The North China Plain therefore requires an evaluation of the need to preserve and possibly strengthen different ESs. The envisioned process of transformation and landscape redesign could be informed by the preferences of stakeholders, especially by those of local farmers. They shape landscapes, influencing cropping systems and ecosystem services, and their benefits and demands must be carefully considered in policymaking (de Snoo et al., 2013).

This paper aims to analyze the perspectives of stakeholders on the importance of ESs provisioning in a case study area in the North China Plain. We use Q methodology to identify perspectives within the stakeholder groups and find the similarity of those perspectives among stakeholder groups. Characteristics associated with farmers' perspectives, as the main stakeholder group, will also be investigated in this study. The ultimate goal is to elucidate stakeholders' perspectives on ESs and to inform policymaking for the redesign and management of agricultural landscapes.

2. Materials and methods

2.1. Case study area

We conducted our study in Quzhou county, located in Heibei Province of the North China Plain (NCP). The surface area of Quzhou is 676.7 km² including 525.2 km² of cropland (NBSC, 2017). Winter wheat, summer maize and cotton are the main crops, occupying together approximately 80 % of the cropland area (NBSC, 2017). The landscapes in Quzhou county are simplified and intensively used for agricultural production, with mostly arable cropping and only a small area of non-crop habitat, and 85.5 % of the population in Quzhou is rural (NBSC, 2017). As a representative area in the NCP, Quzhou has been selected as a National Agriculture Green Development Pilot Demonstration Zone in 2019. Since 2020, the Quzhou government has formulated policies to transform the county towards sustainable agricultural development in the context of the *Agriculture Green Development* (AGD) program, which is facilitated by Chinese national policies (Shen et al., 2020). We chose Quzhou for this study because the area has sustainability issues, and is in need of an agricultural green transition.

Quzhou county has 10 townships including 342 administrative villages. In this study, we selected 35 villages in Quzhou (Fig. S2(a)). Five villages were sampled from each of 7 different village types, which together cover the diversity of socio-economic conditions and agricultural activity within the county (Xu et al., 2024). The sampling strategy was equally weighted among the seven clusters to capture the full diversity of stakeholders' views without regard for the size of each cluster (number of villages in it). Details about the sampled villages and their locations can be found in Cheng et al. (2023), Xu et al. (2024) and Fig. S2.

2.2. Q methodology and data collection

An application of Q methodology proceeds in four steps: (1) research design, (2) data collection, (3) analysis, and (4) interpretation (Zabala et al., 2018). In step 1, the researcher selects a topic or issue to study and describes the discourse around the topic and generates a set of statements (Q set); this set represents different perspectives or opinions on the topic. Here the statements refer to different ESs and the topic is the importance of different ESs delivered by agricultural landscapes in the NCP, and in Quzhou in particular (Section 2.2.1). The researcher then recruits the participants (p set) who are identified as key stakeholders and have knowledge or experience related to the topic (Section 2.2.2). During step 2 (data collection), the researcher conducts the Q sorting,

which involves having participants rank the statements according to their perceived importance (Section 2.2.3). Our Q sort distinguished five levels of importance: very unimportant, unimportant, neutral, important and very important (Fig. 1). The number of statements assigned to each category varied (2–4–8–4–2; 20 statements in total), broadly representing a bell-shaped distribution, with the middle category having the greatest number of "slots" to assign statements (Fig. 1). In step 3 (analysis) the researcher analyzes the data using a variety of multivariate data-reduction techniques (such as principal components analysis or factor analysis) to identify the patterns of Q sorts, representing the main perspectives or views among the stakeholders (Section 2.2.4). Step 4 (interpretation) presents the results to show how the perspectives are different or similar and explains the key characteristics of those perspectives.

2.2.1. Determining the statements for the Q set

Relevant ESs delivered by agricultural landscapes were selected from the four categories of ESs, i.e., provisioning services, supporting services, regulating services and cultural services (Haines-Young and Potchin, 2018; MEA, 2005). We determined 20 relevant ESs of agricultural landscapes in total and for each ES we formulated one statement describing it. Hence, our Q sorting results characterize the perceived importance of different ecosystem services. The statements were initially formulated based on the original definition of ESs that are relevant in the context of agricultural landscapes. Before the formal sorting exercises, three rounds of pilot sorting exercises were done to test whether the statements and pictures were understandable for local farmers and minor revisions were made. The final 20 statements related to 20 ESs (Table 2) were printed onto cards (in Chinese).

2.2.2. Stakeholder identification and selection

Before the survey, we did a stakeholder identification analysis (Reed et al., 2009) to select the most relevant stakeholder groups in Quzhou county. Stakeholder groups were considered relevant if they had basic knowledge about agriculture or potentially had an interest in ESs and agricultural development, or their actions might impact ESs. The final list of stakeholders was established after consulting experts and local informants (Table 3).

We sampled 35 villages across all the 10 townships in Quzhou. The majority of farmers in the villages are smallholders (Xu et al., 2024). Five smallholder farmers were randomly chosen from each village to ensure the survey's feasibility within a time and budget framework. The village heads helped identify up to two leader farmers, defined as those running cooperatives, family farms or having participated in professional trainings. All 35 village heads and nine township heads were

1- Very unimportant	2- Unimportant	3-Neutral	4- Important	5- Very important

Fig. 1. Grids used in the Q methodology sorting exercises with the shape like a normal distribution. The ESs assigned to grids in the columns 1–5 get the corresponding importance scores. The fully filled grids (Q sort) show the preference of all ESs (see a filled example in Fig. S5 of supplementary material).

Table 2

Ecosystem services and the statements used in the Q sorting.

Ecosystem services and categories	Statements
Provisioning services	
Nutrient provisioning	To harvest crops to supply food and nutrients
Profit provisioning	To harvest crops for profit
Regulating services	
Water regulation	To maintain sufficient soil moisture in the farmland
Air quality	Good air quality
Natural hazard regulation	To reduce the yield loss caused by natural hazards
Water quality	Good water quality
Biological control	To reduce pesticide use but use biological control
Pollination	To have effective pollination for the crops requiring pollinators
Climate regulation	Preventing climate warming
Supporting services	
Water provisioning	To have sufficient irrigation water
Soil fertility	Good soil fertility in the farmland
Natural habitats of field border	To have more non-crop greening areas around the farmland
Natural habitats within fields	To have more diversified crops in farmland
Cultural services	
Agricultural tourism	To relax and recreate at rural tourism sites
Scientific and educational value	To get involved in scientific/educational events
Aesthetic value	To have beautiful scenery of the farmland
Belonging value	To have a feeling of belonging living in the rural area
Self-identity value	To have self-identity for farmers
Social relations	To have good social relations within rural communities
Spiritual value	To keep the ancestral graves in the farmland

included (except the township head of Nanliyue who was unavailable). Five citizens were selected randomly from each of the four main townships (Baizhai, Disituan, Huaqiqiao, Anzhai). In these townships, the total cropland area is relatively large, and the townships are distributed in the north, south, west and east of Quzhou County, respectively (Fig. S2).

The county government staff were recruited by a snowball sampling if they had important roles and their departments contributed to the county-level *Agriculture Green Development* program. Companies which were interested in the *Agriculture Green Development* program were selected from a list of locally leading companies and the managers of those companies were recruited. People working in academic institutions and universities with strong relations and cooperation with Quzhou county were also selected by snowball sampling. In total, 327 people finished the Q sorts completely.

2.2.3. Conducting Q sorts and getting importance scores

The Q sorts and interviews were done from December 2020 to April 2021 by a pre-trained team of instructors. We asked the farmers and citizens to sort the cards according to their own opinion, but for other stakeholders, we asked them to represent the point of view of their institution or committee.

During the sorting exercise, each participant was asked to arrange the statements into the grid according to the level of importance of the statements (Fig. 1). For the respondents who had difficulty reading, the instructors read the statements out aloud and explained statements with the help of pictures. In the first round of card-sorting, the participants initially sorted the cards into three piles for “important”, “not important” and “neutral” statements, without a limitation in the number of statements in each pile. During a second round, the participants were asked to decide the extreme answers of “very important” and “very unimportant” statements from the piles of “important” and “unimportant” services in the first round. The remaining statements were placed in the three central columns of the grid ranging between important, neutral and unimportant. The most important and unimportant statements in the Q sorts were then discussed in a post-hoc interview to get a

Table 3

Stakeholder groups and the sample size used in the Q methodology analysis.

Stakeholder group	Code	Explanation	Sample size
Farmers	Farmer	Individuals who are in charge of the crop production in their households. Most farmers are smallholders with farm sizes of less than two hectares. Leader farmers are those who 1) run cooperatives, 2) run family farms or 3) have been trained as professional farmers by the agricultural bureau.	222
Village heads	Village	The village head or key village cadres working on the village committees. These functionaries have a good understanding of the village situation.	35
Township heads	Township	The (deputy) township heads working in the township government take charge of the administrative affairs of rural and agricultural production.	9
County government	County	According to the AGD development planning of local government, the relevant county-level bureaus mainly include the Agriculture and Rural Bureau, Eco-Environment Bureau, Natural Resources and Planning Bureau Development and Reform Bureau, and Water Conservancy Bureau. In each bureau, representatives of the sectors involved in the AGD program were selected.	21
Companies	Company	Leading companies that produce, purchase, or process agricultural (crop) products.	5
Citizens	Citizen	People living in the township centers but are not involved in agricultural activities.	20
Academics	Academic	Researchers from China Agricultural University, local agronomists, and graduate students in Science and Technology Backyards (Zhang et al., 2016). The researchers were often involved in the AGD program together with Quzhou County government.	15

deeper understanding of the participant's reasonings.

2.2.4. Determining the perspectives: Q analysis

The data from the Q sorting were analyzed using Q analysis, which is based on principal component analysis (PCA). Similar to standard PCA, the data are reduced by Q analysis to a few factors (or principal components), which represent the perspectives shared by groups of people. This analysis serves as a tool to uncover the underlying patterns in how people fill the Q sorts similarly, thereby highlighting shared ways of thinking across the group, and revealing the commonalities in how participants rank the statements in the Q sort. We began with a comprehensive Q analysis using all 327 Q sorts. Following this, we conducted separate Q analyses for each of the seven stakeholder groups. This approach allowed us to identify distinct perspectives within each stakeholder group and was deemed necessary due to the over-representation of farmers among the respondents.

For each Q analysis, the number of factors was determined based on the commonly used statistical criteria like scree plots, explained variance, the number of participants flagged and the Kaiser-Guttman criterion (Borcard et al., 2011; Cliff, 1988). These criteria typically indicate the selection of a small number of perspectives. In combination with our expert judgment, we finalized the number of meaningful perspectives, which in this case ranged from two to three (Andreotti et al., 2023).

The factor loadings, z scores and factor scores are the key results from the Q analysis to describe each perspective (Brown, 1980; Zabala et al., 2018) and the calculation is well established in the ‘qmethod’ package (Zabala, 2014) and other Q analysis software. Factor loadings

indicate the relationship between the Q sort and each factor. A high factor loading indicates that the participant's Q sort closely aligns with the pattern of that factor. Participants were assigned, or "flagged" to a factor if their factor loading was significantly high ($p < 0.05$) and the squared loadings for that factor exceeded the sum of the squared loadings for all other factors (Zabala, 2014). The flagged Q sorts are used for subsequent calculations. The z-scores are the weighted averages of the scores given by the flagged Q sorts to a statement. The factor scores of the statements for each factor are derived from z-scores by sorting the statements in ascending order of z-score and assigning the 1–5 scores following the pattern in the Q grid in Fig. 1. This results in a typical and representative Q sort for the factor, which allows easy comparison with other factors (c.f., Yazar and Orth, 2018). A detailed illustration of how to calculate z-scores and factor scores is provided in the supplementary material. We conducted the Q analysis using the rotation method of varimax based on 'qmethod' package (Zabala, 2014) in R version 4.4.0 (R Core Team, 2021).

2.2.5. Comparing preferences for ESs among perspectives

In this study, the statements represent the ESs and the factors denote the different perspectives on the relative importance of these ESs. The factor scores were used to compare the relative importance of individual ESs across perspectives. We also compared the preferences across ES categories by calculating the scaled average factor scores for each ES category. Scaling between 0 and 1 (multiplied by 100, resulting in values 0–100) was needed due to the different number of statements across categories (2 for provisioning ESs, 4 for supporting ESs and 7 for regulating and cultural ESs). We define a perspective prioritizing certain ES categories when the scaled average factor scores were higher than 50. A detailed calculation of scaled average factor scores is provided in the supplementary material.

2.3. Personal, household and village characteristics

Following the Q sort, farmers were asked to fill in a structured questionnaire, which consisted of two main parts: 1) crop production information for each farm, including total planting area and revenues of crops, etc.; 2) basic information of participants, including age, gender, education, years of farming, type of farmer (smallholder farmers or leader farmers), off-farm occupation, family income, etc. Other stakeholders were interviewed to understand their roles and work experience and we collected their age, education level and gender. To protect the privacy of participants, we cannot specify the specific positions of these respondents in their organizations.

2.4. Statistical analysis

The mean and standard deviation of ES preference scores within different stakeholder groups were calculated, and visualized using the "Likert" (Bryer et al., 2016) and "ggplot2" packages (Wickham et al., 2016) in R (R Core Team, 2021).

To test the difference in ES importance scores among stakeholder groups, parameter-free Kruskal–Wallis tests were conducted in R. The confidence level for these tests was 95 %. The mean of each ES importance score from all the stakeholder groups was calculated by dividing by the maximum value, 5 (very important), and plotted as the size in bubble plots by the package of 'ggplot2' (Wickham et al., 2016) using R (R Core Team, 2021).

As farmers were considered a central group of stakeholders, we aimed to establish which farmer characteristics were associated with different perspectives. We expected that personal characteristics, household characteristics and farm characteristics could influence farmer perspectives (Table 4). Kruskal–Wallis tests were conducted for continuous variables to determine whether there were significant differences among perspectives, and Chi-square tests were conducted for categorical variables. The confidence interval for these tests was 95 %.

Table 4

Variables that are potentially associated with perspectives of farmers.

Variables	Units/Categories
Personal characteristics	
Age	years
Education year	years
Experience in agriculture	years
Farmer types	Smallholder farmers, leader farmers
Gender	Female, Male
Occupation other than farming	1) Employment outside of villages 2) Local agricultural employment 3) Local non-agricultural employment 4) Self-employment
Other roles in the village	1) Accountant in the village committee 2) Village doctor 3) Village teacher 4) Village cadres
Household and farm characteristics	
Total owned land area	ha
Non-agricultural income	CNY/year
Agricultural revenue	CNY/year
Main crop type (determined by the most area)	Cereals, Cereals and Fiber, Cereals and Vegetables, Fiber, Vegetables, Others

The analyses were conducted in R (R Core Team, 2021).

3. Results

3.1. Overview of the respondents and ecosystem services importance scores

In total, we obtained 327 completed Q sorts and accompanying interviews. Of the respondents, 13.5 % were females and 86.5 % were males. Respondents were on average 53.7 years of age and had 9.7 years of education (around the age of completing junior high school) (Table S1). More than half of the respondents sorted scientific and educational value, profit provisioning, water provisioning (irrigation), and air quality as important ecosystem services (scores 4 or 5) of agricultural landscapes (Fig. S3). Ecosystem services with low importance (scores 1 or 2) according to more than half of the respondents were the presence of natural habitats (both within the fields and on field borders), agro-tourism, climate regulation, and pollination (Fig. S3).

3.2. Different scorings among stakeholder groups

Stakeholder groups were different in their personal characteristics (Table S1) and their scoring of ecosystem service importance (Table S2 and Fig. 2). Academics, citizens, companies and county governmental staff rated the importance of biological control on average higher than farmers, village heads and township heads did. For cultural services like social relations and spiritual value, citizens had mostly similar importance scores as farmers, while companies cared relatively less, except for scientific and educational values. We found that academics rated the importance of natural habitats in general higher than other respondents. All the stakeholder groups assigned low importance to climate regulation and pollination, and more importance to profit provisioning, air and water quality, and water provisioning.

3.3. Different perspectives within stakeholder groups

As a first step, we conducted a single overarching Q analysis with all 327 participants. This analysis identified three main perspectives: Perspective 1 emphasized the importance of provisioning services; Perspective 2 focused on regulating and supporting services; and Perspective 3 highlighted cultural services (see Table S4 and Fig. S6). This comprehensive Q analysis also illustrated the diversity of perspectives within the stakeholder groups (Table S5). However, because

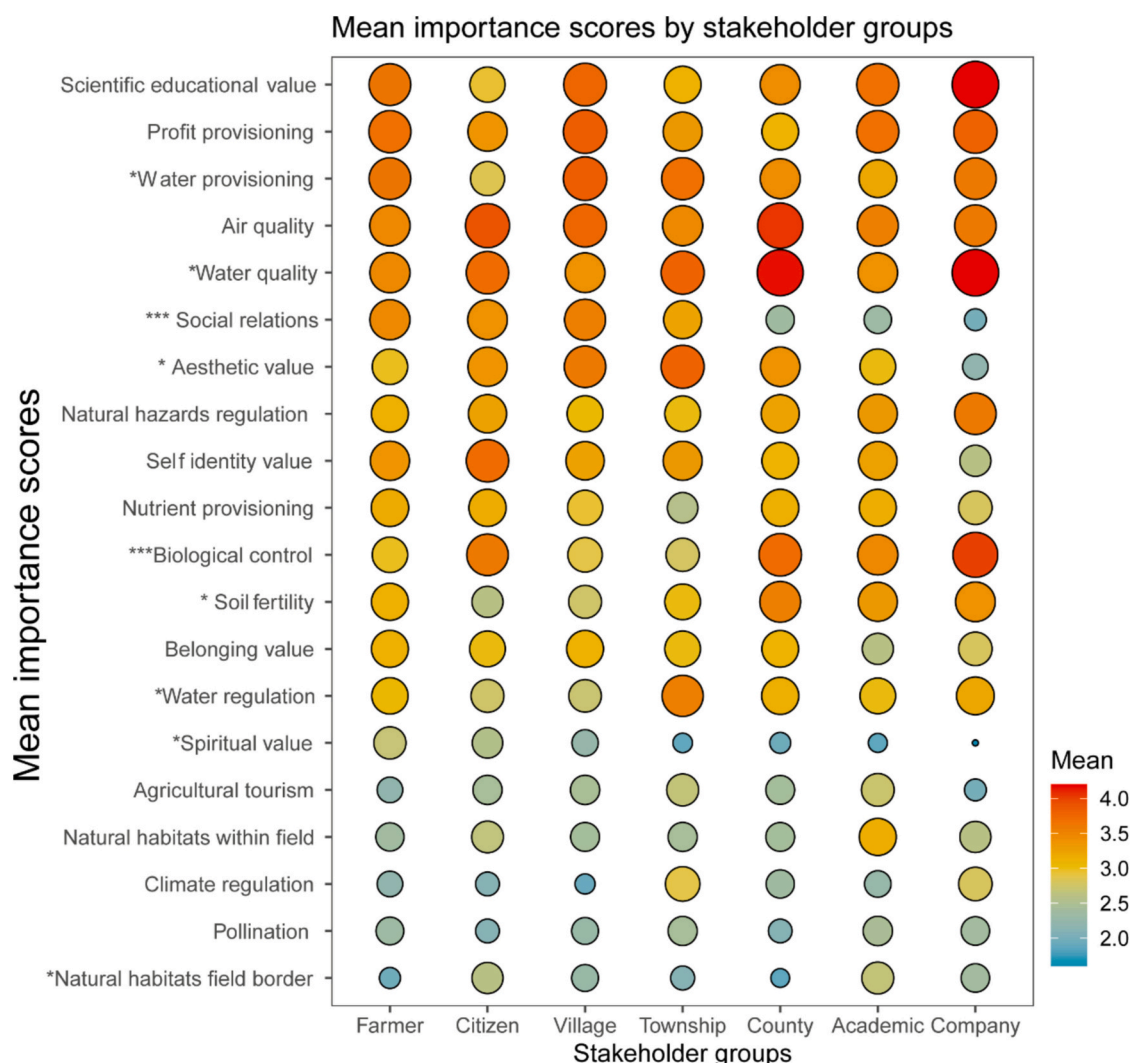


Fig. 2. Means of the ecosystem services importance scores in different stakeholder groups. The color and size of circles shows the absolute value of the importance score (from 1-least important to 5-most important). Significant differences among stakeholder groups are indicated: * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

farmers constitute the majority of participants, the overall analysis may be disproportionately influenced by their views. To address this imbalance and to gain a clearer understanding of how different stakeholder groups prioritize ecosystem services, we performed separate Q analyses for each stakeholder group. This approach allowed for a more nuanced exploration of the specific preferences within each group and provided more targeted insights into their distinct perspectives.

Within each stakeholder group, we found 2 to 3 different perspectives based on the results of Q analysis (Fig. 3 and Table 5). In total, there were 18 different perspectives from 7 different stakeholder groups, reflecting differences in importance assigned to ecosystem services categories (Fig. 3 and Table 5).

Within the group of farmers, the perspective Farmers_1 gave higher importance to provisioning services than other services, the "Nutrients provisioning" and "Profit provisioning" services were assigned the highest importance. The perspective Farmers_2 attributed more importance to regulating services like "Air quality" and "Water quality". The perspective Farmers_3 assigned more importance to cultural services like "Aesthetic value" and "Social relations". Similarly, there were also three different perspectives within the citizens group, with different preferences on regulating (Citizen_2), provisioning (Citizen_3) and cultural services (Citizen_1) (Fig. 3 and Table 5).

There were three different perspectives among village heads. The perspective Village_1 and Village_2 showed similar preference for

provisioning services, but participants sharing the Village_1 perspective gave higher importance to regulating services while Village_2 gave higher importance to cultural services like "Agrotourism" and "Aesthetic value" of agricultural landscapes. Village_3 assigned less importance to provisioning services than other perspectives but gave more importance to supporting services (Fig. 3 and Table 5).

Two different perspectives emerged among academics. The perspective of Academics_1 assigned more importance to supporting services like "Soil Fertility" and "Water provisioning". The perspective of Academics_2 gave more importance to provisioning and cultural services like "Social relations" and "Profit provisioning" (Fig. 3 and Table 5).

Among county government staff, the differences were that County_1 gave higher importance to regulating services ("Air quality" and "Biological control" services) but had the least importance on cultural services. The perspective County_2 gave higher importance to provisioning services, and the perspective County_3 gave higher importance to regulating and cultural services, for example, "Air quality" and "Aesthetic value" services (Fig. 3 and Table 5).

There were two different perspectives among companies, with differences in the preferences mainly reflected in the regulating services, e. g., "Biological control" service. The two perspectives assigned similar importance to provisioning services like "Profit provisioning" service and gave the least importance to cultural services (Fig. 3 and Table 5).

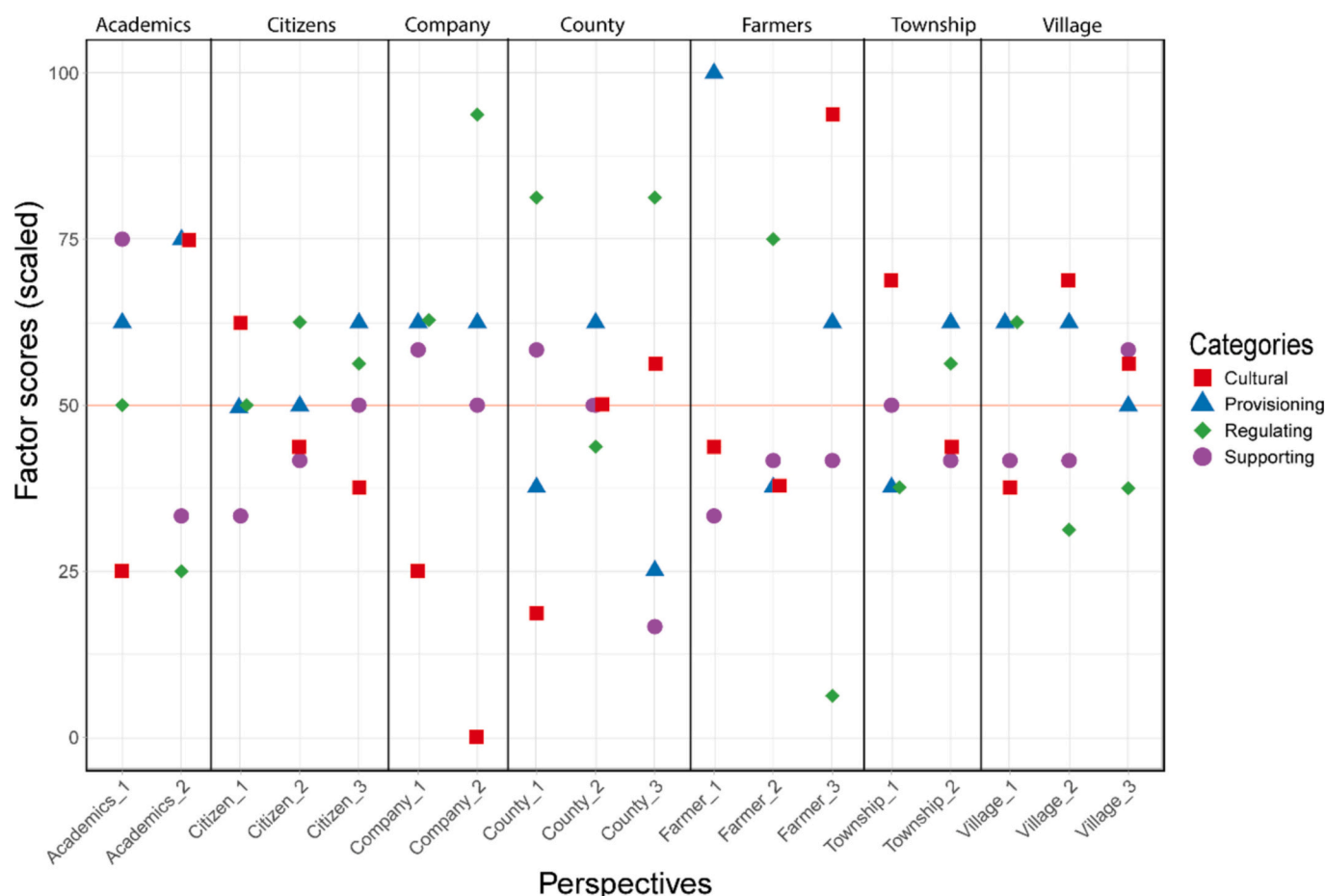


Fig. 3. Different perspectives (x-axis) within each stakeholder group and the average factor scores of each ecosystem service category (y-axis) showing the relative importance of each ES category.

There were also two different perspectives within the group of township heads, with differences in the preferences mainly for cultural and provisioning services. Township_1 gave the least importance to “Nutrients provisioning” service but gave the highest importance to “Agricultural tourism” service, which is quite different from Township_2 (Fig. 3 and Table 5).

3.4. Similar perspectives across stakeholder groups

From Fig. 3 and Fig. S7, we identified 11 perspectives that prioritized provisioning services (i.e. scaled factor scores higher than 50), with representation across all stakeholder groups. Nine perspectives prioritized regulating services. Seven perspectives prioritized cultural services, covering all groups except companies. Lastly, only four perspectives prioritized supporting services, and these were absent among village heads, academics, county government staff, and companies.

Fig. 4 provides a qualitative summary of the prioritization expressed by the identified perspectives. While 11 perspectives prioritize provisioning services, only 2 (Farmer_1 and County_2) show an exclusive prioritization for these services. Other perspectives, such as Citizen_3, Company_1, and Township_2, primarily show a prioritization for provisioning services but also value regulating or supporting services, reflecting a concern for both agricultural production and environmental health (Fig. S7 and Fig. 4).

Among the 5 perspectives that showed a preference for cultural services, only 2 (Citizen_1 and Township_1) showed an exclusive prioritization for this category. Perspectives such as Village_2 and Farmer_3, while primarily showing a prioritization for cultural services,

also indicated a preference for provisioning services (Fig. S7 and Fig. 4).

Similarly, of the 11 perspectives that showed a prioritization for regulating and supporting services, only 3 (Farmer_2, Citizen_2, and County_1) exclusively focused on regulating and supporting services. Perspectives like Village_3 and County_3, while mainly prioritizing regulating or supporting services, also showed a preference for cultural services (Fig. S7 and Fig. 4).

Perspectives that were broadly similar in some instances still differed in the importance assigned to specific services (Table 5). For example, the perspectives of Farmer_1 and County_2 both agreed that profit provisioning is very important, but County_2 gave lower importance to human dietary nutrition provisioning than Farmer_1. Although perspectives Company_2 and Academics_1 had a similar prioritization for the environmental dimension of ESs and provisioning services, they have opposite ranks on the importance of climate regulation service.

3.5. Farmer attributes related to the different perspectives

Among the farmer participants, we found three different perspectives (Fig. 3). The farmers with the perspective Farmer_1 (prioritizing provisioning services and therefore labeled ‘Production’ perspective) were on average older farmers with more experience working on agricultural production but lower education (Fig. 5 (a), (b) and (d)). We also found that the farmers with the perspective of Farmer_1 tended to have smaller farm areas and lower incomes (Fig. 5 (c), (e) and (f)), and had less non-agricultural income compared with farmers with other perspectives. Farmers with the perspective Farmer_2 (prioritizing regulating services and therefore labeled “Environment” perspective) were on average younger and had received more education than farmers in the Farmer_1

Table 5

Factor scores of each perspective within stakeholder groups based on Q analyses.

Perspectives	Provisioning services		Regulating services							Supporting services				Cultural services						
	Nutrient provisioning	Profit provisioning	Water regulation	Air quality	Natural hazards regulation	Water quality	Biological control	Pollination	Climate regulation	Water provisioning	Soil fertility	Natural habitats of field border	Natural habitats within fields	Agricultural tourism	Scientific and educational value	Aesthetic value	Belonging value	Self-identity value	Social relations	Spiritual value
Farmer_1	5	5	3	3	4	3	3	2	2	4	3	1	2	1	4	2	3	3	4	3
Farmer_2	2	3	3	5	4	5	3	3	2	3	4	2	2	1	4	3	3	3	4	1
Farmer_3	3	4	2	3	2	3	2	1	1	3	3	2	3	3	4	5	4	4	5	3
Academics_1	3	4	3	3	3	4	4	3	1	5	5	2	3	2	4	3	2	3	2	1
Academics_2	3	5	2	2	3	3	3	2	2	3	3	3	1	1	4	3	4	4	5	4
County_1	3	2	4	5	3	4	5	2	3	3	4	3	3	1	4	3	3	2	2	1
County_2	3	4	3	3	3	5	2	2	2	5	3	1	3	3	4	4	3	4	2	1
County_3	2	2	3	5	4	5	3	2	4	3	3	1	1	2	3	4	4	3	3	3
Village_1	3	4	3	4	4	5	3	2	2	5	3	1	2	1	4	3	3	3	3	2
Village_2	3	4	3	4	2	3	3	2	1	4	2	3	2	5	4	5	3	3	3	1
Village_3	3	3	2	3	3	3	4	3	1	4	2	4	3	1	2	5	3	4	5	2
Citizen_1	3	3	3	5	3	4	4	1	1	3	2	2	3	3	2	4	2	5	3	4
Citizen_2	3	3	3	4	4	4	4	2	2	3	3	2	3	1	2	3	5	5	3	1
Citizen_3	4	3	3	5	4	3	3	2	2	3	3	4	2	1	5	3	2	3	4	1
Company_1	3	4	3	3	3	4	5	2	3	4	4	2	3	2	5	2	3	3	1	1
Company_2	3	4	3	4	4	5	4	3	5	3	3	3	3	1	2	1	2	2	2	3
Township_1	1	4	3	3	2	4	2	3	2	4	3	2	3	5	4	5	3	3	3	1
Township_2	3	4	4	3	3	3	3	3	3	5	3	1	2	1	2	4	4	5	2	2

and Farmer_3 groups. Farmers with the perspective Farmer_3 (prioritizing cultural and to a lesser extent provisioning services and labeled “Culture” perspective) included more females and more leader farmers (Fig. S4 (a) and (b)). The group of farmers having an environment perspective (Farmers_2) had diverse roles in the villages and a variety of non-farming occupations besides farming (Fig. S4 (d) and (f)).

4. Discussion

This study analyzed the perspectives of stakeholders on the importance of ecosystem services (ESs) provisioning in a case study area in the North China Plain using Q methodology. It introduces a novel application of Q methodology by directly linking the statements to specific ESs, differentiating our work from traditional approaches. In general, the participants assigned the highest importance to provisioning services and the lowest importance to supporting services in this study. We found most of the respondents regarded scientific and educational value, profit provisioning, water provisioning (irrigation), and air quality as important ecosystem services of agricultural landscapes in Quzhou (Fig. S3 in Supplementary Material). Notable differences emerged among stakeholder groups, e.g., academics assigned higher importance to supporting services while the companies assigned less importance to cultural services (Fig. 2). There were different perspectives within each stakeholder group (Fig. 3), and totally we found 18 different perspectives from 7 different stakeholder groups. While most perspectives showed a combination of preferences for at least two different ES categories, we also observed similarities across stakeholder groups (Fig. 4 and S7). Five perspectives mainly prioritized provisioning services, seven perspectives prioritized regulating services and supporting services, and four perspectives prioritized cultural services. This pattern is further confirmed by the overarching single Q analysis (Fig. S6 in Supplementary Material). As the main stakeholder group, farmers’ perspectives were associated with personal characteristics (age and education, Fig. 5), and with farm and household characteristics (farm size and income, Fig. 5).

4.1. Importance of agroecosystem services: Consensus and disagreement

In this study, the perspective of prioritizing agricultural production (provisioning services) was held by most of the participants, and this perspective was found in all of the stakeholder groups. Increasing the profit of crop production and supplying sufficient dietary nutrients are also local goals for the long term in the policy agenda. Similar to previous studies, especially those conducted in areas undergoing rapid economic development, we found that provisioning services were generally preferred over regulating, cultural and supporting services in agricultural landscapes (Wang and Zhao, 2023; Zhang et al., 2020). Moreover, there was a consensus among stakeholder groups and participants with different perspectives that irrigation water provisioning, water and air quality, and science and educational value are important ESs in Quzhou. The interviewed stakeholders assigned a low importance to preventing climate warming although they were aware that temperature is increasing (Cheng et al., 2023). Farmers may also not be aware that agricultural activities and practices could contribute to climate warming (Madhuri Sharma, 2020) and they stated that they cannot control how nature changes. A previous study in China found that farmers have developed adaptive strategies to deal with the impact of climate change like increasing irrigation (Zhai et al., 2018). It suggests that they do not want to suffer the damage from climate change, but they are also passive about preventing climate warming since they consider it something outside their control.

Pursuing mainly provisioning services may decrease the delivery of cultural services in agricultural landscapes (Landis, 2017). Prioritization of cultural services was a minority perspective in this study. People holding this perspective may face more conflicting interests with the majority. As a specific example in the North China Plain, tombs in the agricultural landscape, while hindering mechanical operations, hold significant cultural value. Removing them for land consolidation would be resisted by farmers, particularly older farmers. However, the farmers, whose cultural services are most affected, often have the least power in decision-making processes (Hanaček and Rodríguez-Labajos, 2018).

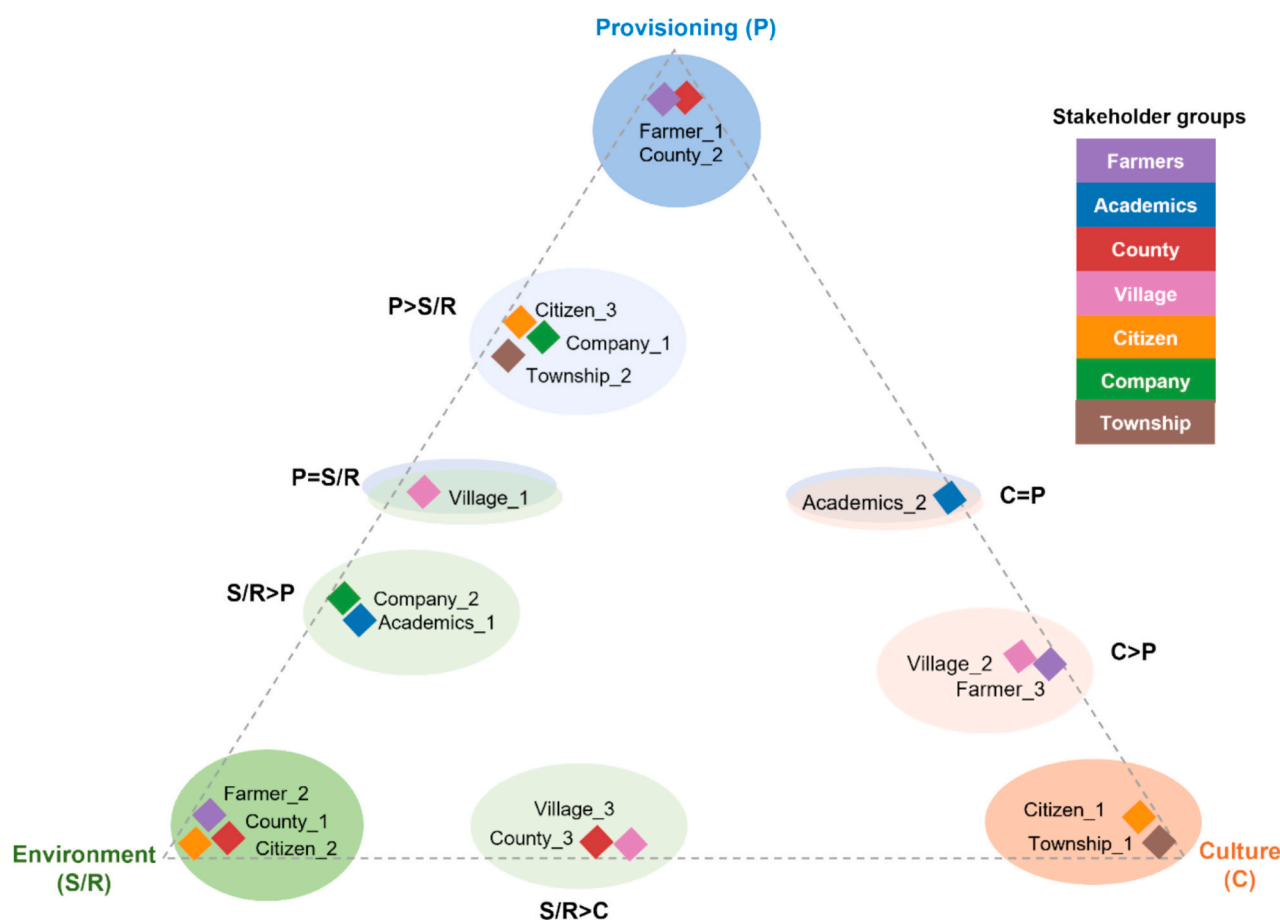


Fig. 4. Qualitative illustration of the similarity of perspectives based on ES category preferences. The points in the figure represent the perspectives, with different colors indicating the stakeholder groups. The ellipses, also in different colors, represent the ecosystem service (ES) categories that are the priority for certain perspectives. Points located at the vertices of the triangle indicate perspectives that show an exclusive priority for the corresponding ES category, represented by a darker color. Points positioned along the line connecting two vertices show that these perspectives prioritize both ES categories (indicated by a lighter color), but one is relatively more preferred. For example, “C > P” means that both cultural and provisioning services scored >50 for the scaled average score, but cultural services are preferred over provisioning services by stakeholders with the perspectives Village_2 and Farmer_3. S/R indicates the largest value assigned for either supporting or regulating services.

Thus, it is important for policy or decision-makers to realize such power asymmetry among the stakeholders while making decisions for land use change and planning, instead of only focusing on major perspectives.

Certain patterns reveal ecosystem services demand bundles: e.g., urban visitors favored cultural services (e.g., aesthetics, leisure), while rural farmers preferred provisioning services (e.g., agricultural production, timber) (Zoderer et al., 2019). Our study also demonstrated that people hold different perspectives showing a preference for different (combinations of) ES categories. Like previous studies, we found that within the population of farmers, the perspectives towards ESs were quite diverse (Teixeira et al., 2018). We observed differences in preferences among stakeholder groups. However, unlike a previous study (Zhang et al., 2020) where experts prioritized regulating services and farmers prioritized provisioning services in a typical Karst basin, our study found both farmers and experts placed greater importance on provisioning services. The possible reason is that the case study area is a county that mostly relies on the rural population and agricultural production and is located in the main crop production area.

Gender, age, and education have often been found to be important factors in understanding people’s perceptions of ESs (Lima and Bastos, 2019). In this study, we provided further evidence similar to previous studies that the farmers who prioritized agroecosystem regulating and supporting services were younger (Maas et al., 2021), higher-educated or larger-scale farm farmers (Teixeira et al., 2018). We found no evidence that farmers’ perspectives have a statistically significant

association with gender which could be expected since the number of female participants was limited. To promote *Agriculture Green Development* in the North China Plain, recommendations or interventions should be tailored to farmers with diverse perspectives. For example, for older farmers who prioritize traditional provisioning services, it is essential to focus on improving economic returns from current cropping systems while preserving cultural traditions. In contrast, younger, more educated farmers are more likely to support transitioning to ecological and multifunctional cropping systems and landscapes. This shift requires targeted professional training in modern agronomic technologies, along with strong policy support from government bodies and research institutions.

4.2. Different perspectives of academics

Integrating semi-natural habitats could support beneficial insect populations and stimulate the provisioning of ESs like biological control and pollination in intensively used, simplified agricultural landscapes like those of Quzhou (Holland et al., 2017). We found that, however, except for the academics, stakeholders assigned less importance to supporting services associated with maintaining semi-natural habitats within fields or on the field borders. Interestingly, still more than 30 % of the respondents, especially some non-farming stakeholder groups such as citizens, county governmental staff and companies found the biological control service important. Similar to previous studies (e.g., Maas

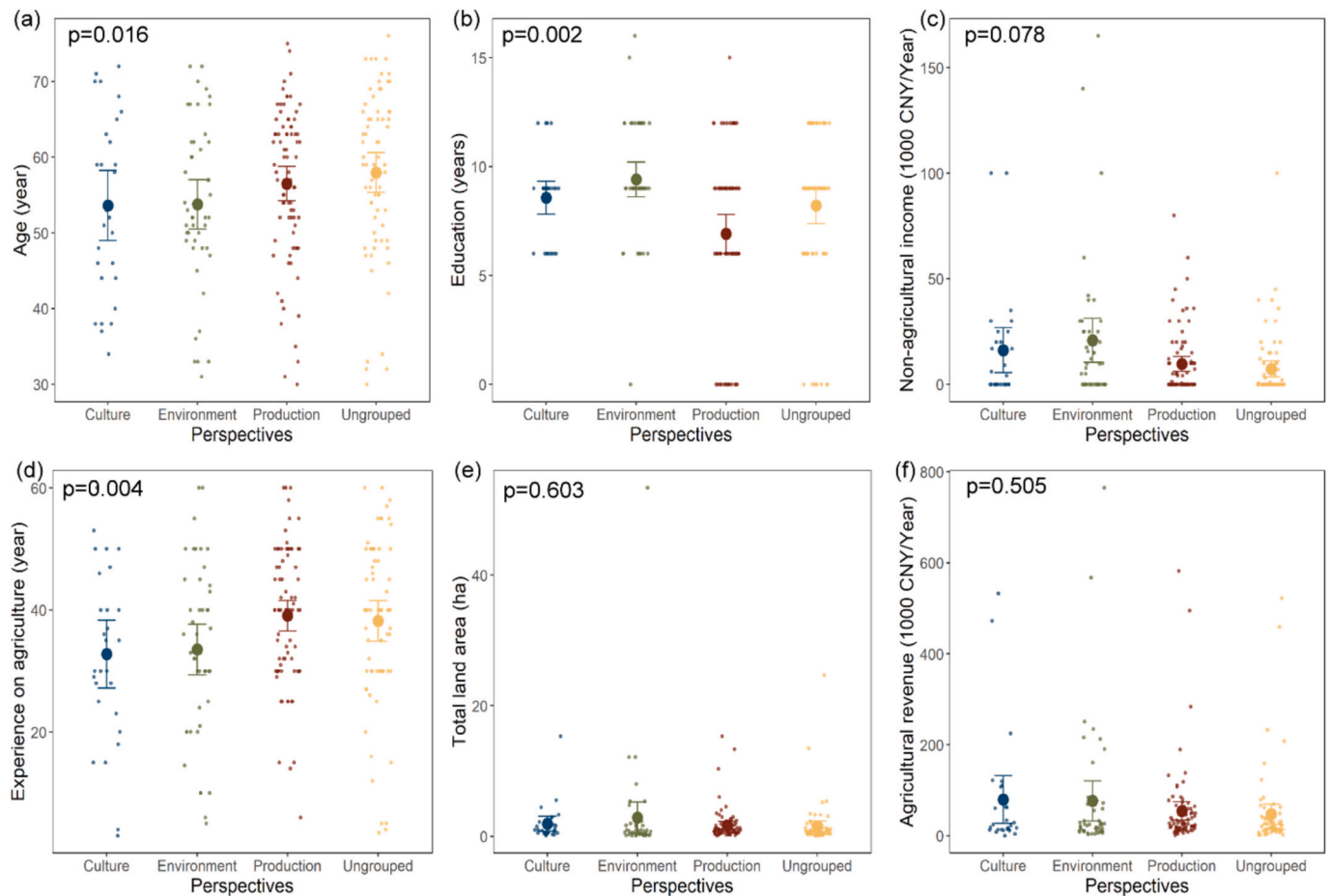


Fig. 5. Personal and household/farm characteristics of farmers for each perspective. The x-axis represents in each panel the four different perspectives of farmers. Production perspective (Farmer_1) indicates a priority for provisioning services; Environment perspective (Farmer_2) indicates a priority for regulating services; Culture perspective (Farmer_3) indicates a priority for cultural and provisioning services. Here we define a perspective prioritizing certain ESs categories when the scale average scores were higher than 50. Farmers not assigned to any specific perspective are categorized as “Ungrouped”. Significant differences in characteristics among farmer groups are indicated: * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

et al., 2021), we found that scientists tended to give more importance to natural habitats and related ESs than farmers.

However, the questions would be: even if developing more semi-natural habitats in simplified agricultural landscapes is necessary, why do people underestimate the importance of semi-natural habitats, and what will be the potential roadblocks for developing more semi-natural habitats? Currently, people perceive natural habitats as causing difficulties in mechanical operations and negative effects on the yield for the crops by the field edges where the windbreaks or hedgerows grow. Lacking knowledge and understanding regarding the relationship between semi-natural habitats and ecosystem services like biological control makes that people do not acknowledge the benefits of semi-natural habitats, and farmers would perceive even mainly the disadvantages (Salliou and Barnaud, 2017). Importantly, the perceived costs and potential yield losses associated with establishing more semi-natural habitats discourage many stakeholders, especially farmers, as there is a lack of supportive policies or subsidies in place at the local level (Pe'er et al., 2017).

4.3. Limitations

We attempted in this study to cover different types of stakeholders (stakeholder groups), however, citizens were underrepresented. Due to time and budget constraints, citizen sampling might lack precision, not encompassing the full spatial and typological spectrum. Despite the female farmers' involvement in agricultural production, their decision-

making independence is often lower than that of males (Ge et al., 2023), and some of them did not want to represent their households to participate in the survey so we interviewed relatively few females in this study. The perception of female and male farmers could be different, for example, females tend to assign more importance to regulating services and sustainability than males (Mponela et al., 2023). Since female perspectives in decision-making were often neglected (Rola-Rubzen et al., 2020), studying the divergent demands of female and male farmers on the importance of ESs is pivotal in future studies for advancing agricultural sustainability with a more gender-sensitive approach.

Unlike traditional Q methodology with small samples of participants, our large sample case showed that a large proportion of the farmers were not assigned to certain perspectives (Fig. 5). The formulation of the statement did not engage with stakeholders. The elicited perspectives thus might not cover all possible viewpoints existing in the whole population in reality (Sneegas et al., 2021). Nevertheless, Q methodology was found to be useful for capturing and understanding the different perspectives on ESs importance in this study. It stimulated expressing opinions when people hesitated to speak openly about a certain topic. Based on our experience, stakeholders' thoughts were more deliberate while they participated in Q sorting and they could rank the ESs importance with gradients.

This study analyzed certain factors influencing people's preference for ecosystem services. However, the preferences can be influenced by many factors other than personal, farm and household characteristics used in this study. Some intrinsic factors or psychological factors, e.g.,

belief, personality, worldview, and awareness, were not included in this study, but they may also affect people's preferences for ESs (Peter et al., 2022).

Quzhou County is a prototypical agricultural production zone within the North China Plain, where stakeholder perspectives are predominantly centered on production functionality. This specific focus raises questions regarding the external validity of our findings, i.e., the ability to generalize our findings in a study to other locations, times, and participants (Roe and Just, 2009). To address this, future research should extend to other socio-ecological regions, allowing for a comparative analysis of our results against those from other contexts to better understand the broader applicability and potential limitations of Q methodology, which could also ensure a more comprehensive understanding of the preferences for multiple ESs beyond agricultural productivity.

Internal validity is concerned with the believability of the results (Streiner, 2020). In Q methodology studies, internal validity can refer to the transparency in selecting statements and participants, following recommended steps to minimize subjectivity, etc. (Zabala et al., 2018), which is what we have done in our study. We also conducted several analyses to probe into the robustness of our results, but we did not specifically validate the results through additional data collection within the study area. Future studies could explore ways to further validate the results of Q methodology. For example, one approach, as suggested by Webler et al. (2009), involves having participants self-validate the perspectives identified after the Q analysis. In that regard, focus group discussions could be valuable for validating patterns of perspectives within specific groups, and discussing potential conflicts and areas of consensus (Cuppen et al., 2010). While such mixed approaches seem promising for a deeper understanding of the perspectives elicited by Q methodology, they also pose some practical limitations. First and foremost, analyzing the data, constructing the perspectives, and visualizing the results all take time, so it would almost certainly require a follow-up meeting at a much later point with the participants. Post-hoc interviews with participants seem more feasible and may also help to gain deeper insights into the reasoning behind their perspectives, though it would not be possible to directly discuss the perspectives with the participants.

4.4. Implications and outlook

Our study contributes to the literature that used Q methodology to map consensus and dissensus on ES provision. Previous studies have employed similar techniques to rank the importance of ESs in various contexts, such as in wetland ecosystems (Mahlalela et al., 2022), coastal lagoon ecosystems (Sy et al., 2018), wildlife management (Bredin et al., 2015), but this study gives the first evidence in the context of intensively managed agricultural landscapes. Methodologically, this study innovated by incorporating a visual approach during Q sorting practices, which enhances participants' comprehension of the statements. Additionally, we adopted a multiple-step approach to compare the perspectives from diverse stakeholder groups, ensuring a comprehensive understanding of the varying views on the importance of ESs. We anticipate that these methodological advancements expand the applicability of Q methodology in agricultural science.

It has been well recognized that a "One-size-fits-all" policy would not meet the different demands for landscape management from stakeholders. This analysis aims to contribute to integrating the stakeholders' preferences into future agricultural landscape redesigning and planning aimed at enhancing multifunctionality. The Q methodology results can assist the process in two ways: 1) engaging the stakeholders while understanding the difference of their perspectives and 2) setting clear prioritized goals based on stakeholders' perspectives.

Integrating stakeholders' participation is one of the key elements for the landscape approach to improve the sustainability in landscapes with land uses that result in conflicting outcomes (Sayer et al., 2013). Although the diversity of the perspectives provides a challenge for policymaking, the results from Q methodology can give a better

comprehension of what the different perspectives are about. Policy-makers can tailor communication strategies to engage stakeholders with distinct views effectively (Maas et al., 2021). Coupled with other possible participatory methods, Q methodology can also be used to monitor the changes in stakeholders' demands along the projects' progress (Andreotti et al., 2023). However, it is necessary for stakeholders to understand concretely how their decision will impact the landscape and thus their livelihoods and well-being. The outcomes and trade-offs of the ecosystem services supply from agricultural landscapes need to be well-informed for stakeholders in different policy scenarios (Bernues et al., 2019). We also suggest that future studies complement Q methodology with deep interviews or focus groups for participants to self-validate their perspectives.

To co-design farming systems and landscapes beyond economic goals, diverse stakeholder preferences necessitate additional objectives (Meynard et al., 2012). With people's preferences for ESs, decision-makers could identify the essential action points in the next step to improve the ESs with higher importance and allocate resources efficiently for maintaining human well-being and agricultural landscape sustainability (Smith and Sullivan, 2014). Multi-criteria Decision Analysis (MCDA) techniques could be used to prioritize ecosystem services based on stakeholders' preferences, such as Analytic Network Process (ANP) method (Parra-López et al., 2008; 2009). These techniques can weigh indicators based on stakeholders' preferences for ecosystem services to translate the multitude of science-based indicators into a more tangible set of indicators, which may better align with the needs and perceptions of nonscientist stakeholders such as citizens, resource managers, and policymakers (Groot and Rossing, 2011). The other way around, stakeholders' preferences can also be translated into specific science-based indicators for long-term monitoring and informing ESs management. The MCDA methodologies can further be used to calculate the priorities of each individual ecosystem service based on the scores from Q methodology in this study. It will be useful for future decision-making if the preferences for ESs can be linked with measurable indicators or management and policy options or alternatives (Parra-López et al., 2008). However, it is also worth noting that the multi-criteria decision-making approach can never give the "right answer" for decision-making, it is only to inform the decision-making process and to provide a structured way of thinking (Belton and Stewart, 2002). To determine the real priorities for enhancing ESs in certain areas, participatory decision-making with multiple stakeholders for certain projects is necessary.

5. Conclusions

This study revealed that while most stakeholders considered provisioning services as the most important, they also recognized the significance of non-provisioning services such as water quality, air quality, and scientific and educational value. The perspectives on the relative importance of ecosystem services (ESs) varied significantly both among and within stakeholder groups. Generally, stakeholders' preferences showed a combination of at least two ES categories, with notable similarities across stakeholder groups. Five perspectives primarily showed the preferences for provisioning services, seven showed the preferences for regulating and supporting services, and four showed the preferences for cultural services. Although the existence of diverse and complex perspectives makes it challenging to find a unified goal for all stakeholders, our research helps identify similar perspectives among different stakeholders, which can be a foundational basis for future collaboration and setting specific goals.

The findings on the relationship between farmers' preferences and characteristics highlight the importance of targeted interventions and policies that enhance agroecosystem services in line with farmers' preferences. Specifically, there should be an emphasis on traditional provisioning services for older farmers, while prioritizing non-provisioning services for younger generations. However, scientists

must also remain mindful of potential conflicts and challenges when promoting regulating and supporting services.

Based on a case study, we applied an adapted Q methodology mainly to show its abilities to uncover diverse stakeholder perspectives. Awareness of different perspectives within and among different stakeholder groups is the first step to facilitating open and effective communication while co-designing for more sustainable and multifunctional agricultural landscapes. In future agricultural landscape planning and redesigning projects, we hope Q methodology can also aid the establishment of clear, prioritized objectives for agricultural landscape multifunctional development in different socioecological contexts.

CRedit authorship contribution statement

Jiali Cheng: Writing – review & editing, Writing – original draft, Visualization, Investigation, Formal analysis, Data curation. **Andries Richter:** Writing – review & editing, Supervision, Conceptualization. **Wen-Feng Cong:** Writing – review & editing, Supervision, Resources, Funding acquisition. **Zhan Xu:** Investigation, Data curation. **Zhengyuan Liang:** Investigation, Data curation. **Chaochun Zhang:** Supervision, Resources. **Fusuo Zhang:** Supervision, Resources, Project administration. **Wopke van der Werf:** Writing – review & editing, Supervision, Conceptualization. **Jeroen C.J. Groot:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

None.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.agry.2024.104187>.

Data availability

Data and R script used in the analyses are available from the corresponding authors on request.

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